MILITARY SPECIFICATION

WIRE, INSULATED, HARD DRAWN ALLOY OF COPPER-CADMIUM, NO. 23 AWG

This specification has been approved by the Naval Ordnance Systems Command, Department of the Navy

1. SCOPE

1.1 Scope. This specification describes a special purpose hard drawn alloy of copper-cadmium conductor insulated with an ionometric type polyolefin for underwater use. Hereinafter, the insulated No. 23 AWG conductor shall be referred to as "wire."

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposals form a part of this specification to the extent specified herein.

SPECIFICATION

Military

MIL-P-46124	Plastic Molding and Extrusion Material Ionomer Resin
MIL-W-82598	Wire, Copper-Cadmium Alloy

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-202	Test Methods for Electronic and Electrical Component Parts

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC- 6145

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

E 8 Tension Test

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19013.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Description. The wire described by this specification consists of a solid hard drawn alloy of copper-cadmium conductor insulated with an ionometric type polyolefin applied in a single extrusion.

3.2 General requirements

3.2.1 Preproduction sample. Unless otherwise specified in the contract or order, a preproduction sample of the wire shall be manufactured using the methods and procedures proposed for the production. The sample will be tested as specified in Section 4 herein and is for the purpose of determining, prior to production, that the contractor's production methods are capable of yielding wire that complies with the technical requirements of the contract.

3.2.2 Material. Unless otherwise specified in the contract, requisition, or order, all materials shall conform to the requirements listed below.

3.2.2.1 Conductor. The conductor shall be AWG #23 hard drawn alloy of copper, in accordance with MIL-W-82598. The conductor shall be of uniform quality throughout. The surface shall be free from oils, greases, soaps, detergents, or other foreign matter before insulation is applied.

3.2.2.2 Insulation. The insulation shall consist of extruded Ionometric Type Polyolefin, Type V, Class 1, in accordance with MIL-P-46124. The insulation shall be free from splinters, blisters, and other nonhomogeneities visible to the normal eye. There shall be no repairs or patches in the insulation. The insulation thickness shall be 0.008 inch minimum. The insulation shall be natural in color, which is transparent and slightly tan to colorless. (See 6.3).

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3.2.3 Dimensions. Wire lengths will be specified in the contract, requisition, or order. Other dimensions shall conform to Table I.

TABLE I

Normal Conductor Size #23 AWG	Diameter of Insulated Conductor (Inches)		
	Min	Nom	Max
0.0226 ±0.0002 Inch	0.039	0.041	0.043

3.2.4 Marking. Each length of wire shall be assigned a designated serial number by the contractor immediately after the application of insulation. This number shall identify the length of wire throughout production, testing, and shipping. Each reel shall have two identification tags, one secured to the end of the wire and the other secured to the outside of the reel. Each tag shall contain the following information:

(a) Manufacturer's Name
(b) Type of Insulation
(c) Specification of Wire
(d) AWG Size of Conductor
(e) Purchase Order Number
(f) Lot Number
(g) Serial Number
(h) Length
(i) Conductor Resistance/1000' Corrected to +20 degrees C (+68 degrees F)
(j) Insulation Resistance/1000'

3.2.5 Conflicting requirements. In the event of conflict between the documents referenced herein and the contents of this specification, the contents of this specification shall apply.

3.3 Processing requirement

3.3.1 Wire twist. No "flyer" type applicators shall be used at any stage of the wire fabrication. The wire shall be applied to or removed from a reel by rotating the reel itself.

3.4 Performance requirements. The wire shall meet the following performance requirements:

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3.4.1 Direct current resistance. The dc resistance shall be 27 ohms maximum per 1000 feet of wire when corrected to +20 degrees C (+68 degrees F) and tested in accordance with 4.5.1.

3.4.2 Spark. The wire shall withstand, without breakdown, the dry spark test voltage specified in 4.5.6.

3.4.3 Dielectric strength. The wire shall withstand, without breakdown, the dielectric strength test specified in 4.5.4.

3.4.4 Insulation resistance. The wire shall exhibit an insulation resistance of no less than 20,000 megohms per 1000 feet when tested in accordance with 4.5.5.

3.4.5 Tensile strength. The wire shall withstand a tensile force of 33 pounds minimum when tested in accordance with 4.5.3.

3.5 Workmanship. The workmanship shall be first class in every respect and uniform throughout each lot of wire offered for delivery. The standards of workmanship exhibited in any approved preproduction sample, subject to any qualification stated in the Government's notice of approval, shall be determinative of the requirements of the contract relative to workmanship insofar as not specifically covered by applicable specifications.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The inspection and testing of the wire shall be classified as follows:

4.2.1 Preproduction inspection

4.2.2 Quality conformance inspection

4.3 Sampling. Samples taken for the purpose of the tests prescribed in this specification shall be representative of the material furnished to fulfill a particular contract or order.

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4.3.1 Preproduction sampling. A representative sample of four wire lengths shall be manufactured using the methods and procedures proposed for the production material. This sample of wire shall be tested as specified in 4.5 for the purposes of determining compliance with the requirements of this specification. Further production of the item by the contractor, prior to the approval of the contracting activity or completion of the tests on the preproduction sample, shall be at the contractor's risk.

4.3.2 Quality conformance sampling. Each length of wire submitted for acceptance shall be tested for conductor resistance, dielectric strength, insulation resistance, and dry spark test as specified in 4.5. For visual, dimensional and tensile properties, a representative sample shall be selected from each lot of material in accordance with MIL-STD-105, Inspection Level II, with an acceptable quantity level (AQL) of 2.5 percent. The representative samples shall be obtained from a point not less than 10 feet from the outside end of the reel. The samples selected for visual and dimensional examination may be the same as those selected for tensile strength, but shall be evaluated separately.

4.3.2.1 Lot size. A lot shall consist of the quantity of the specified wire lengths that are insulated during one operator's normal working shift. Any subsequent testing, respooling, or cutting shall have no effect on the lot size.

4.3.2.2 Rejections. If the number of nonconforming sample units in the sample exceeds the acceptance number specified in 4.3.2 for that sample size, the entire lot or reel shall be rejected subject to the provisions of the section on "Acceptance and Rejection" on MIL-STD-105.

4.4 Test conditions. Unless otherwise specified the wire shall be tested under the following conditions:

(a) Temperature: +59 degrees F (+15 degrees C) to +95 degrees F (+35 degrees C).

(b) Altitude: Local atmospheric.

(c) Humidity: 90 percent relative, maximum.

4.5 Inspection procedures. Wire selected in accordance with 4.3 shall be tested for acceptance in accordance with the following:

4.5.1 Conductor resistance. The direct current resistance of each reel of wire shall be measured in accordance with Method 303 of MIL-STD-202, and shall be as specified in 3.4.1. This measured resistance shall be recorded on the identification tag.

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4.5.2 Visual and dimensional inspection. A wire sample shall be checked dimensionally for conformance to the requirements of this specification. A micrometer caliper or an instrument of equal accuracy shall be used to determine finished wire diameter. Insulation wall thickness shall be determined by use of an optical instrument, such as a toolmaker's microscope of sufficient accuracy to determine thickness within ±0.0002 inch.

4.5.3 Tensile strength. A wire sample shall be tested in accordance with ASTM E 8. The snubbing type grip shall be used. Breaking shall be considered rupture of the insulation or conductor, whichever occurs first. The free lengths of wire between the snubbing sheaves shall be no less than 6 inches. The minimum breaking strength is specified in 3.4.5.

4.5.4 Dielectric strength. Each length of wire shall be wound onto a reel and the entire length of wire, except approximately 1 foot of each end, shall be immersed in tap water at a temperature not to exceed +30 degrees C (+86 degrees F). At the end of 4 hours of soaking, and while still in the bath, the wire shall be tested by applying a potential difference of 1000 volts rms at commercial power frequency between the conductor and the surrounding water. This potential shall be applied for not less than 1 minute and any breakdown shall constitute a failure.

4.5.5 Insulation resistance. Upon completion of 4.5.4, and while the reel is still immersed, insulation resistance measurements shall be made with a dc potential of 500 volts. The insulation resistance of the wire shall be measured in accordance with MIL-STD-202, Method 302, Test Condition B, and shall be greater than 20,000 megohms per 1000 feet.

4.5.6 Dry spark. A dry spark test shall consist of passing the wire through a chain-electrode spark testing device which subjects the insulation to an alternating potential of at least 4000 volts rms at 3000 Hz for at least 0.003 second, or at least 4000 volts rms at commercial power frequency for a period of not less than 0.20 second, or a potential of at least 5000 volts dc for a period of not less than 0.01 second. Any flashover shall constitute a failure.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, and packing

5.1.1 Cleaning. Not applicable.

5.1.2 Preservation application. Not applicable.

5.1.3 Supply reels. The reels used to ship the wire may be fiber glass, metal, or wood or combinations in order of preference. There shall be no burrs, sharp edges, splinters, or other hazards to the wire. The reels shall

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have an outside flange diameter of 22 inches maximum, with a drum diameter of 10 inches minimum; a length of 12-3/4 inches maximum, with a traverse of 10 ±1 inches; and an arbor hole of 3 ±1/4 inches. Each flange shall have a 3/4 ±1/32 inch diameter break hole or "dog hole" 3/4 inch deep (minimum) on a 3-1/2 ±1/32 inch radius from the center of the reel.

5.1.3.1 Reel winding. The wire must be wound uniformly at 7 pounds tension minimum, and at approximately 1/4 inch traverse advance per revolution of the reel. The inside end shall have a free length of at least 72 inches measured from the reel center, and shall be protected by insulating sleeving or grommets where it passes through the drum and/or flange of the reel. The sleeving shall extend for at least 3 inches on either side of each point of contact with the drum or flange. The inside end shall be coiled into a diameter of not less than 4 inches. The running end of the wire must be securely fastened to the reel flange to prevent loosening.

5.1.4 Unit packaging and packing. Wire reels shall be packaged to be acceptable to common carriers for transportation at the lowest rate applicable to ensure adequate protection against damage during shipment and storage. The wire on the reel must be protected by a heavy chipboard or similar material between the reel flanges. The chipboard must overlap at least 1/8 of the circumference over the wire. Reels must be shipped securely fastened to pallets. No more than six reels may be placed on any one pallet. Reels must be positioned on the pallets such that the edges of the flanges on any reel cannot come into contact with the chipboard on any other reel either on the same or adjacent pallet.

6. NOTES

6.1 Intended use. The wire is intended for use as a portion of the communication circuit between certain ordnance equipments.

6.2 Ordering data. The contract or order shall specify the following:

(a) The specific title, number, and date of this specification.
(b) Length of wire, or multiples thereof, per reel.
(c) Exceptions to this specification.
(d) The applicable drawings and other documents.
(e) Disposition of the samples.
(f) Responsibility for additional samples and inspection to replace samples which fail.
(g) Method of shipment.
(h) Provisions for returnable reels.

6.3 Insulation material. A product that has been developed and tested to meet the requirements of MIL-P-46124(MR), Type V, Class 1, and has been used

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successfully as the insulation material on the wire described in this specification, is E. I. DuPont DeNemours and Company's Surlyn A 1800. This paragraph is in no way to limit other products that meet the requirements of MIL-P-46124(MR), Type V, Class 1, from being used, provided all other requirements of this specification are adhered to.

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Preparing Activity
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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION MIL-W-82599(OS) Wire, Insulated, Hard Drawn Alloy of Copper-Cadmium, No. 23 AWG

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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